

CyberWell RAG

Final Evaluation Report

Corpus	25 CyberWell PDF Reports (2022–2026)
db	Chroma-db (local)
llm	Gemini 2.5 flash
Embedding Model	Gemini embedding 001
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1. System Architecture & Chunking

Pipeline

```
data/raw/*.pdf → load & chunk → embed (Gemini) → ChromaDB (cosine) →  
retrieve(top-k) → generate_answer (Gemini Flash) → answer()
```

Corpus Properties

Property	Value
Documents	25 PDFs
Format	PDF (text via PyMuPDF)
Question types	Factual, numerical, temporal, comparison, negation
Coverage	Platform policy, trend alerts, annual reports, case studies on online antisemitism (2022–2026)

Chunking Strategy

Primary configuration: paragraph_window with 2000-word budget and 200-word paragraph overlap. Chunk IDs stored as {doc_id}_chunk_{NNNN} in data/processed/chunks.jsonl.

- paragraph_window: Packs whole paragraphs until the word budget; overlap rewinds by paragraphs.
- fixed_words (ablation): Sliding word windows — shorter windows align numerical/temporal gold facts more reliably.

Embedding & Vector Index

Component	Choice
Embeddings	gemini-embedding-001 (Vertex AI)
Task types	RETRIEVAL_QUERY / RETRIEVAL_DOCUMENT
Vector DB	ChromaDB (persistent, cosine HNSW)
Retrieval	Dense vector search, top-k = 5, similarity = $1/(1+\text{distance})$

Prompt Design

- Answer only from context; fixed refusal phrase if context is missing.
- Inline citations [chunk_id]; emphasize exact numbers/dates; clear yes/no for negation.
- Temperature: 0.2. Entry point: src/generation.py.

2. Evaluation Results

Retrieval Metrics — paragraph_window, k=5

Gold set: eval/gold_set.jsonl (100 questions, 20 per category). Results file: eval/results/retrieval_results_paragraph_k5.json (2026-05-24).

Metric	Value
Hit@5	90.0% (90/100)
MRR	0.73
Recall@5	90.0%
Retrieval misses	10 / 100

Hit@5 by Question Category

Category	n	Hit@5	MRR
Factual	20	100%	0.81
Temporal	20	90.0%	0.85
Numerical	20	90.0%	0.79
Comparison	20	85.0%	0.58
Negation / Absence	20	85.0%	0.65

Weakest categories: Comparison and Negation/Absence (85.0% Hit@5) — cross-report and October 7-themed queries frequently pull trend alerts or related annual reports instead of the gold chunk.

Manual Answer Evaluation (n=15, primary config)

Stratified sample: 3 questions per category. Labels in eval/answer_labels.jsonl per eval/ANSWER_EVAL_RUBRIC.md.

Label	Count	Rate
Correct	13	86.7%
Partially correct	2	13.3%
Incorrect	0	0%
Unsupported / Hallucinated	0	0%

Answer accuracy (strict): 86.7% (13/15). Partial credit (correct + 1/2 partial): 93.3%.

3. Ablation Study & Failure Analysis

Ablation Table

All runs: 2000-word / 200-word overlap, gemini-embedding-001. Each strategy evaluated against its index-aligned gold set.

Experiment	Hit@5	MRR	Misses	Notes
paragraph_window, k=5 ★	90.0%	0.73	10/100	Primary
fixed_words, k=5	96.0%	0.85	4/100	Retrieval ablation
paragraph_window, k=1	62.0%*	0.62	38/100	Hit@1 ablation
fixed_words, k=1	76.0%*	0.76	24/100	Hit@1 ablation

* Hit@1 for k=1 rows. Answer accuracy measured on 15-question stratified sample for primary config only.

Key takeaway: fixed_words retrieval at k=5 reaches 96.0% Hit@5 vs 90.0% for paragraph_window (+6.0 pp). Dropping to k=1 costs ~28 pp (paragraph) or ~20 pp (fixed); fixed_words retains a clearer advantage at k=1.

Failure Analysis (10 misses at k=5)

- Semantic collision across reports:** Bondi Meta policy on Conspiratorial Self-Victimisation — gold chunk_0000 displaced by denial/CSV report chunks on the same theme.
- Statistic in a different chunk:** AI dataset "98.4% from June 2025" — gold is chunk_0000 but retrieval returns chunk_0005–0002 from the same PDF.
- Wrong report in top-k:** Ye #TheNoticing mention count — gold Ye alert chunk_0000 displaced by 2022 annual report and Twitter insights.
- October 7 report family confusion:** Negation on November 2023 war trends — gold war-trends chunk_0000 displaced by October 7 denial trend alerts.
- Annual-report vs trend-alert mix-up:** 2023 cross-platform removal rate — gold annual_report chunk_0000 displaced by 2024 annual, evolution pre/post Oct 7, or war-trends chunks.

Recommended Mitigations

- Hybrid retrieval (BM25 + dense) for exact statistics and report titles.
- Reranking over top-20 before generation.
- Smaller chunk sizes (300–700 words) for short factual QA.
- Metadata filters (report type, date) for trend-alert vs annual-report disambiguation.
- Expand manual answer audit beyond the initial 15-question sample.
- Citation validation against retrieved chunk text.